UNIVERSITY OF NORTH CAROLINA
CHARLOTTE

University of North Carolina at Charlotte
Department of Mathematics and Statistics
Math 2164 - Test 2: Matrix Algebra, Determinants & Vector Spaces – Version (a)
(75 minutes)
Fall 2026

Name: Student ID:

Instructions: Show all your work. If your work does not fit inside a rectangle, continue it on the back of the page, write down every detail there, and clearly label which problem and part it belongs to. When you do this, write *"continued on back"* inside the rectangle so that the work on the back is graded.

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1. Let $T, S : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformations

$$T(x, y) = (2x + y,\ x - 3y), \qquad S(x, y) = (x + 2y,\ -y).$$

(a) Find the standard matrices A_T and A_S.

(b) Find the standard matrix of the composition $S \circ T$.

2. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}.$$

(a) Use the Gauss-Jordan algorithm on $[A \mid I_3]$ to compute A^{-1}. State clearly each row operation you perform.

(b) Use A^{-1} to solve the system

$$\begin{cases} x_1 + x_2 = 2 \\ x_2 + 2x_3 = 3 \\ x_1 + x_2 + x_3 = 4 \end{cases}$$

3. (a) Compute the determinant using a cofactor expansion:

$$\det \begin{pmatrix} 3 & 0 & -2 \\ 1 & 2 & 4 \\ -1 & 1 & 0 \end{pmatrix}.$$

(b) Use row operations and the determinant of a triangular matrix to compute

$$\det \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & -2 \\ 0 & 2 & 8 & -1 \\ 0 & 0 & 0 & -2 \end{pmatrix}.$$

4. Let A and B be 3×3 matrices with $\det(A) = -2$ and $\det(B) = 5$. Compute each of the following.

(a) $\det(3A)$

(b) $\det(A^{-1}B^T)$

(c) $\det(A^3B^{-1})$

5. Use Cramer's Rule to solve the system

$$\begin{cases} 3x_1 + 4x_2 = 2 \\ x_1 + 2x_2 = -1 \end{cases}$$

6. Determine whether each set H is a subspace of the given vector space. Justify your answer by checking the subspace properties, or by giving a specific counterexample.

(a) $H = \{p \in P_2 : p(0) = p(2)\}$, as a subset of P_2.

(b) $H = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 3\}$, as a subset of $\mathbb{R}^3$.

7. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $T(\vec{x}) = A\vec{x}$, and let

$$A = \begin{pmatrix} 1 & 2 & -4 & 3 \\ 2 & 3 & -5 & 4 \\ 1 & 1 & -1 & 1 \end{pmatrix}.$$

Set

$$B = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and assume $B \sim A$.

(a) Compute a basis for $\mathrm{Ker}(T)$.

(b) Compute a basis for $\mathrm{Col}(A)$.

(c) State $\mathrm{rank}(A)$ and $\mathrm{nullity}(A)$, and verify that they satisfy the Rank Theorem.

8. Consider the polynomials in P_2

$$p_1(t) = 1 + t, \quad p_2(t) = 2t + t^2, \quad p_3(t) = 2 - 4t - 3t^2.$$

(a) Is $\{p_1, p_2, p_3\}$ linearly independent? If it is linearly dependent, find a linear dependence relation.

(b) Find a basis for $H = \text{Span}\{p_1, p_2, p_3\}$, and state $\dim H$.

9. Determine whether each statement is true or false. No justification required.

(a) If A is an $n \times n$ matrix, then $\det(-A) = -\det(A)$.

◯ True
◯ False

(b) If A is an invertible $n \times n$ matrix, then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

◯ True
◯ False

(c) If the columns of a 4×4 matrix A are linearly dependent, then $\det(A) = 0$.

◯ True
◯ False

(d) If A is a 6×4 matrix with $\text{nullity}(A) = 1$, then $\text{rank}(A) = 3$.

◯ True
◯ False

(e) Any set of three vectors in $\mathbb{R}^4$ spans $\mathbb{R}^4$.

◯ True
◯ False

(f) If A is row reduced to B, then $\text{Col}(A) = \text{Col}(B)$.

◯ True
◯ False